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Mitigation of thermoacoustic instability utilizing steady air injection near the flame anchoring zone

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ABSTRACT

The objective of this work is to investigate the effectiveness of steady air injection near the flame anchoring zone in suppressing thermoacoustic instabilities driven by flame–vortex interaction mechanism. We perform a systematic experimental study which involves using two different configurations of air injection in an atmospheric pressure backward-facing step combustor. The first configuration utilizes a row of micro-diameter holes allowing for air injection in the cross-stream direction just upstream of the step. The second configuration utilizes an array of micro-diameter holes located on the face of the step, allowing for air injection in the streamwise direction. The effects of each of these configurations are analyzed to determine which one is more effective in suppressing thermoacoustic instabilities at different operating conditions. The tests are conducted while varying the equivalence ratio and the inlet temperature. The secondary air temperature is always the same as the inlet temperature. We used pure propane or propane/hydrogen mixtures as fuels. Combustion dynamics are explored through simultaneous pressure and heat release-rate measurements, and high-speed video images. When the equivalence ratio of the reactant mixture is high, it causes the flame to flashback towards the inlet channel. When air is injected in the cross-stream direction, the flame anchors slightly upstream of the step, which suppresses the instability. When air is injected in the streamwise direction near the edge of step, thermoacoustic instability could be eliminated at an optimum secondary air flow rate, which depends on the operating conditions. When effective, the streamwise air injection prevents the shedding of an unsteady vortex, thus eliminating the flame–vortex interaction mechanism and resulting in a compact, stable flame to form near the step.

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1. Introduction

A phenomenon often observed in continuous combustion systems is self- sustained pressure and flow oscillations forming as a result of the resonant interactions between unsteady heat release mechanisms and the acoustic modes of the system, referred to as the thermoacoustic instability [1,2]. Thermoacoustic instability occurs under lean burn conditions, where most emissions and efficiency benefits are achieved, or near stoichiometry, where high power density is the objective. Thermoacoustic instability is undesirable since it may cause flame extinction, structural vibration, flame flashback and even structural damage. Several mechanisms appear to be present in a combustor that instigate the unsteady heat release, including: flame–acoustic wave interactions, flame–vortex interactions, equivalence ratio oscillations, flame wall interactions and unsteady stretch rate, all of which may be present individually or simultaneously [3–5], and hence promoting thermoacoustic instability.

Active combustion control strategies that modulates the fuel flow rate [6–13], or air flow rate [14] can be applied to suppress thermoacoustic instabilities in premixed combustors by disrupting the coupling mechanisms that support these instabilities. While effective in suppressing the instabilities, these approaches require high speed actuators and add significant complexity to the design of the combustors. For this reason, developing simple, passive control strategies that directly impact the flame anchoring zone with minimal complexity are desirable.

The potential for suppressing thermoacoustic instability by injecting steady air [11,15,16], or by injecting open-loop, low-frequency modulated air [17] near the flame anchoring zone has been shown in the past. In these approaches, the secondary air flow is thought to change the velocity field and disrupt the flame–vortex interaction, which has been shown to be the primary instability mechanism in dump combustors [18]. In this paper, we inject steady air flow near the flame anchoring zone through choked micro-diameter holes. The small hole diameter allows for large velocities with low air flow rates. We perform a systematic experimental study with extensive range of operating conditions, which involves using two different configurations of secondary air injection in an

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atmospheric pressure backward-facing step combustor. The first configuration is a row of microjets allowing for air injection in the cross-stream direction just upstream of the step. The second configuration is an array of microjets located on the face of the step, allowing for air injection in the streamwise direction. The effects of each of these configurations are analyzed to determine which one is more effective in suppressing thermoacoustic instabilities at different operating conditions. The tests are conducted while varying the equivalence ratio and the inlet temperature. The secondary air temperature is always the same as the inlet temperature. Previous studies have shown that enriching hydrocarbon fuels with small amounts of hydrogen significantly reduces the equivalence ratio at the lean blowout limit of the combustor [19–22]. For this reason, we add small amounts of hydrogen to the primary fuel, in this case propane. In order to understand the role of air injection in suppressing the instabilities, we record the flame images using a high speed video camera and investigate the effect of secondary air injection on combustion dynamics.

2. Combustor description

Fig. 1 shows a schematic diagram of the backward-facing step combustor. The combustor consists of a rectangular stainless steel duct with a cross section 40 mm high and 160 mm wide. The air inlet to the combustor is choked. 0.45 m downstream from the choke plate, a 0.15 m long ramp contracts the channel height from 40 mm to 20 mm followed by a 0.4 m long constant-area section that ends with a sudden expansion back to 40 mm. The step height is 20 mm. The overall length of the combustor is 5.0 m. A circular exhaust pipe comprises the last 3.0 m of the combustor. The combustor is equipped with quartz viewing windows. An air compressor supplies air up to 110 g/s at 883 kPa. A pair of Sierra C100M mass flow controllers allow arbitrary propane/H_2 mixtures at maximum flow rates of 2.36 g/s for propane and 0.30 g/s for hydrogen. Fuel is injected through several spanwise holes in a manifold located 93 cm upstream of the step. At this fuel injection location, equivalence ratio oscillations are not present and the instability is solely driven by the flame–vortex interaction mechanism [18]. Images of the flame are captured using an image-intensified Phantom v7.1 high-speed camera and used to examine the combustion dynamics. An optical bandpass filter centered at 430 nm is placed in front of a Hamamatsu H9306-02 photosensor module to measure the integrated CH^* chemiluminescence emitted by the flame, which is proportional to the instantaneous heat-release rate [23,24]. Pressure measurements are obtained using Kulite high intensity microphones designed for laboratory investigations. The inlet and secondary air temperatures are adjusted using Osram Sylvania 18 kW and 4 kW in-line electric heaters, respectively, with on/off temperature controller. All data are acquired using a National Instruments PCIe-6259 data acquisition board and the Matlab Data Acquisition Toolbox. A custom Matlab code is used to store the data and control the experiment.

Secondary air can be injected from 12 and 0.5 mm diameter choked holes drilled along the combustor width in the cross-stream direction 5 mm upstream of the step, or in the streamwise direction near the corner of the step. These two air injection configurations are termed respectively the *normal* microjets and the *axial* microjets. Their positions are shown schematically in Fig. 2. Since the flow from the microjets is choked, it does not couple with the acoustics of the combustor allowing for steady flow rates.

3. Results

By performing a parametric study we determined the optimum secondary air flow rate required to suppress the instabilities using both normal and axial microjets. By changing the equivalence ratio, inlet temperature, secondary air temperature and fuel composition we compare the effectiveness of microjet air injection using each injection configuration at different operating conditions. Using high-speed video images, we captured the combustion dynamics at each microjet configuration at distinct operating modes. We compare the pressure amplitudes under different operating conditions and microjet flow rates. We report the pressure amplitude in terms of the overall sound pressure level (OASPL) using the pressure sensor located 1.28 m downstream of the choke plate, which is near the location of the flame, unless otherwise is mentioned. The OASPL in dB is defined as:

$$\mathrm{OASPL} = 10\ \log_{10}\overline{\left[\frac{p(t)-\overline{p(t)}}{p_o}\right]^2} \tag{1}$$

where overbars indicate average values, $p(t)$ is the pressure measured in an interval $t_1 < t < t_2$ and $p_o = 2 \times 10^{-5}$ Pa. The Reynolds number is fixed at 6500, which is based on the step height, and the primary air flow rate.

3.1. Combustion dynamics without passive control (baseline)

We investigated the stability map and combustion dynamics by systematically varying the operating conditions without passive control. Fig. 3 shows the OASPL as a function of the equivalence ratio at inlet temperatures of $T_{in} = 300$ K and 600 K, using pure propane or propane enriched with 50% by volume hydrogen (4.4% by mass; 10.6% by LHV) as the fuel.

Examining Fig. 3 in detail, for $T_{in} = 300$ K, we observe three distinct operating bands independent of the fuel composition. The band near the lean blowout limit, with an OASPL of around 130 dB, is the stable band, where small vortices are shed from

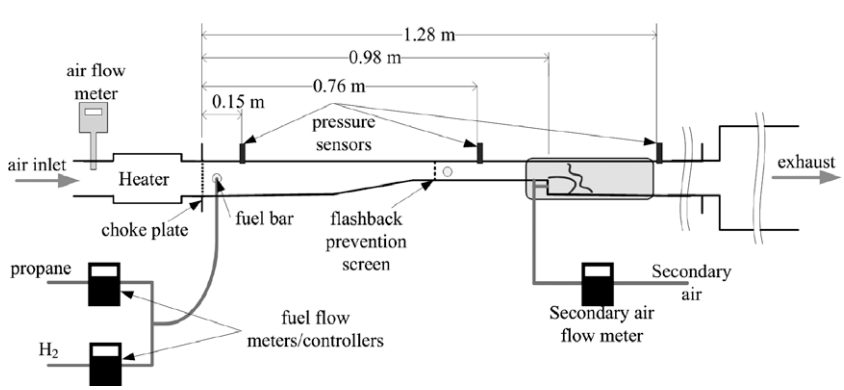

Fig. 1. The schematic diagram of the experimental combustor setup.

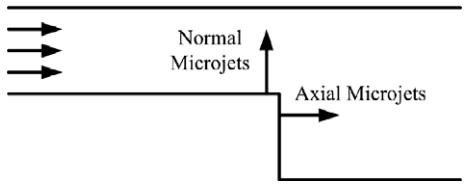

Fig. 2. Schematic diagram showing the microjet air injection configurations with the fuel/air mixture entering from the left.

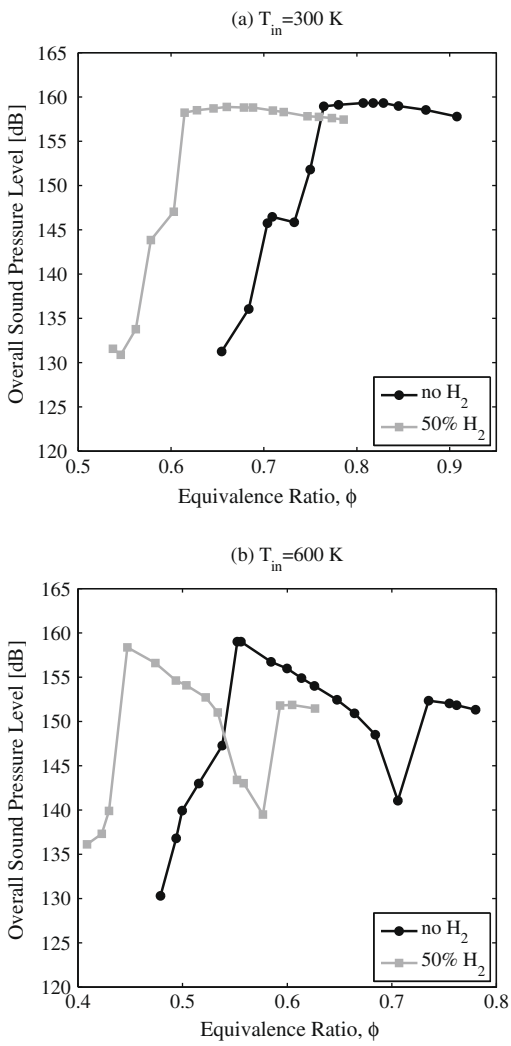

Fig. 3. OASPL as a function of equivalence ratio at different fuel compositions and inlet temperatures.

the step at a high frequency which does not correspond to the acoustic modes of the combustor, and thus there is no coupling between heat-release rate and pressure. The band corresponding to the OASPL of around 145 dB is the quasi-stable band where heat release oscillations are weak. The band around 160 dB is the unstable band showing strong unsteady interactions between the flame and the wake vortex causing large heat-release rate oscillations and loud burning. The resonant frequency is 36 Hz. When $T_{in} = 600$ K, we observe an additional operating band at high equivalence ratios corresponding to OASPL levels of around 152 dB, occurring after a sharp increase in the OASPL. In this operating band the combustor is still unstable; however, the system resonates at 132 Hz and is referred to as the high-frequency unstable mode. Compared to the unstable mode, the wake vortex in this case remains closer to the step as a result of the less time available in an unstable cycle for it to convect downstream. The stability map shifts towards lower equivalence ratios with hydrogen enrichment and higher inlet temperature. For more details, see Ref. [18].

3.2. Combustor acoustics

In Fig. 4, we plot the pressure modeshapes along the length of the combustor for typical unstable, high-frequency unstable and quasi-stable modes. The sign of the rms pressure amplitude is reversed whenever measurements at two consecutive locations are out of phase. Fig. 4 clearly shows that the quasi-stable and the unstable modes correspond to the 1/4 wavemode, and the high-frequency unstable mode corresponds to the 5/4 wavemode of the combustor. The pressure oscillations do not satisfy the open boundary condition at the end of the exhaust tube (at 5.0 m), suggesting a longer acoustic length. It is also interesting to see that the 3/4 wavemode is not excited.

In order to explain the observed resonant frequencies, we analytically modeled the combustor acoustics by solving the linear one-dimensional wave equation, which explained in detail in Refs. [25,18]. Fig. 5 shows the modeshapes and associated resonant frequencies corresponding to the 1/4, 3/4, 5/4 and 7/4 wavemodes of the combustor obtained from the analytical model. The observed resonant frequencies as well as the modeshapes are predicted well using the acoustic model. The pressure nodes of the 5/4 wavemode occur around 0.8 m and 2.9 m, which matches closely the experimental measurements shown in Fig. 4. These results clearly prove that the unstable and high-frequency unstable operating modes of the combustor correspond to the 1/4 and 5/4 wavemodes, respectively. The figure also shows that the location of the flame is very close to a pressure node in the 3/4 wavemode, where the pressure amplitude is very low. This is the reason why the 3/4 wavemode was not excited.

Next, we investigate the combustion dynamics in the same combustor while introducing choked microjets near the flame anchoring location. The objective of using microjet air injection is to modify the hydrodynamics of the flame anchoring zone, since the instability is governed by the large wake vortex shed from the step.

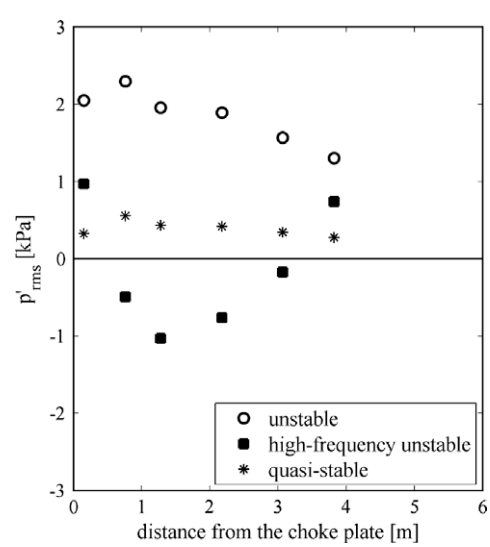

Fig. 4. Pressure modeshapes corresponding to the unstable, high-frequency unstable and quasi-stable operating modes.

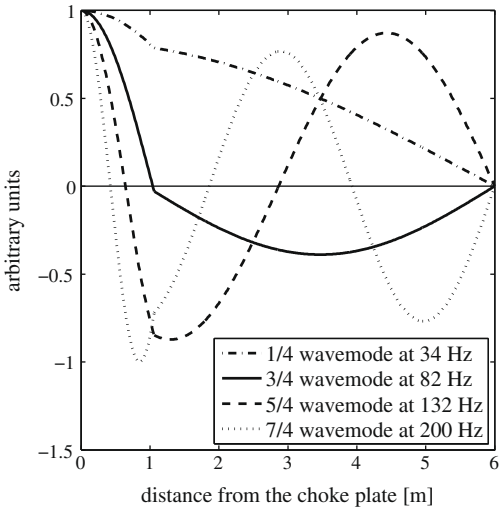

Fig. 5. Pressure modeshapes calculated from the acoustic model corresponding to the 1/4, 3/4, 5/4 and 7/4 wavemodes.

3.3. Combustion dynamics with normal microjets

We performed experiments to investigate the effect of secondary air injection through normal microjets on the combustion dynamics. We measured the pressure amplitude and investigated the flame images at different fuel compositions, inlet temperatures, equivalence ratios and microjet flow rates.

3.3.1. Pressure measurements

In Fig. 6, we plot the OASPL as a function of the equivalence ratio based on the main air flow rate at different normal microjet flow rates without hydrogen enrichment and with 50% by volume hydrogen enrichment. The temperatures of the inlet mixture and the secondary air flow are 300 K. The main air flow rate is approximately 17.5 g/s. Without hydrogen enrichment, normal microjets destabilize the unstable dynamics ($\phi > 0.77$), causing near 4 dB increase in the OASPL. The quasi-stable dynamics is stabilized with normal microjets, reducing the OASPL by a maximum of 12 dB to the values corresponding to the stable mode. The stable mode is not impacted by normal microjets. When 50% by volume hydrogen is added to propane, there is reduction in the OASPL compared to the baseline when $\phi > 0.68$. At equivalence ratio of 0.80, this reduction is near 18 dB, the OASPL is close to the values observed in the quasi-stable regime. When the equivalence ratio is reduced below 0.68, the normal microjets destabilize the unstable dynamics further, similar to the cases without hydrogen enrichment. Again, normal microjets stabilize the quasi-stable dynamics and do not impact the stable dynamics with 50% by volume hydrogen enrichment.

In order to investigate the effect of normal microjets further, we plot the combustor's frequency response in Fig. 7 corresponding to the cases in Fig. 6. These figures show the amplitude of pressure oscillations as a function of the frequency for the ranges of equivalence ratios in Fig. 6. High amplitudes correspond to darker regions. Without hydrogen enrichment, there is no change in the resonant frequency when normal microjets are introduced. The resonant frequency is around 40 Hz in all cases and corresponds to the 1/4 wavemode of the combustor. With 50% by volume hydrogen enrichment, when the equivalence ratio is greater than 0.76, the OASPL is very low which corresponds to the plateau observed in Fig. 6b. The dominant frequencies in these cases are 120 Hz and 175 Hz. At these operating conditions the normal microjets are able to suppress the dynamics.

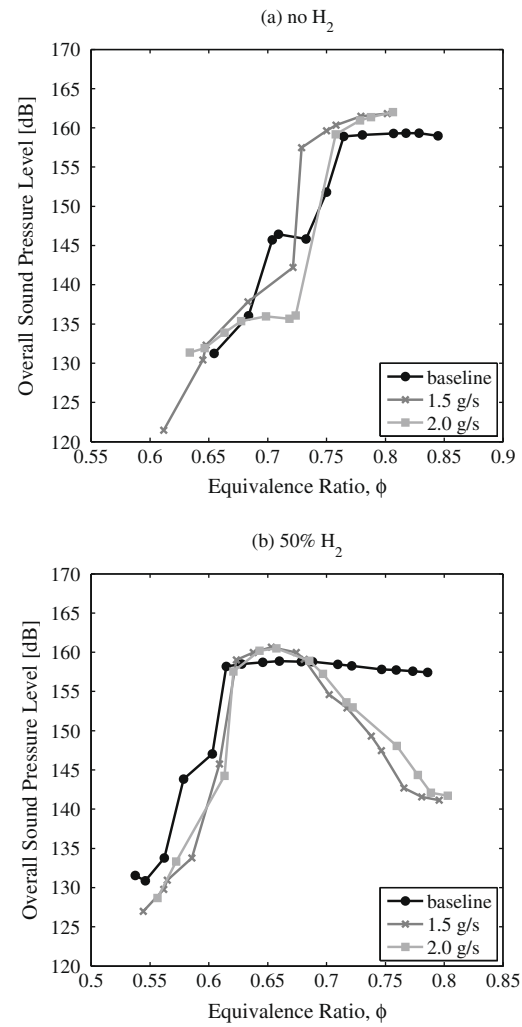

Fig. 6. OASPL as a function of equivalence ratio at different normal microjet flow rates (a) without hydrogen enrichment and (b) with 50% by volume hydrogen enrichment. The temperatures of the inlet mixture and the normal microjets are 300 K.

Next, we increase the temperatures of the inlet mixture and the secondary air flow to 600 K. The main air flow rate in this case is around 28.5 g/s. Fig. 8 shows the OASPL as a function of the equivalence ratio based on the main air flow rate at different normal microjet flow rates without hydrogen enrichment or with 50% by volume hydrogen enrichment. As the equivalence ratio increases from its value where the transition to quasi-stable operating mode takes place, i.e. $\phi \sim 0.55$ without hydrogen enrichment and $\phi \sim 0.45$ with 50% by volume hydrogen enrichment, first the normal microjets destabilize the unstable dynamics further, then the OASPL drops to the values corresponding to the quasi-stable dynamics just before suddenly increasing to higher values, which suggests excitation of a higher acoustic mode. The equivalence ratio at the transition to the higher acoustic mode is lower with normal microjets compared to the baseline. Fig. 8 also shows that the normal microjets stabilize the dynamics in the quasi-stable regime and have no impact on the dynamics in the stable regime similar to the behavior observed in Fig. 6.

We plot the combustor's frequency response in Fig. 9 corresponding to the cases in Fig. 8. Without normal microjets, at higher equivalence ratios, the acoustic mode around 132 Hz is excited. This acoustic mode corresponds to the 5/4 wavemode of the combustor as shown in Section 3.2. However, with the normal micro-

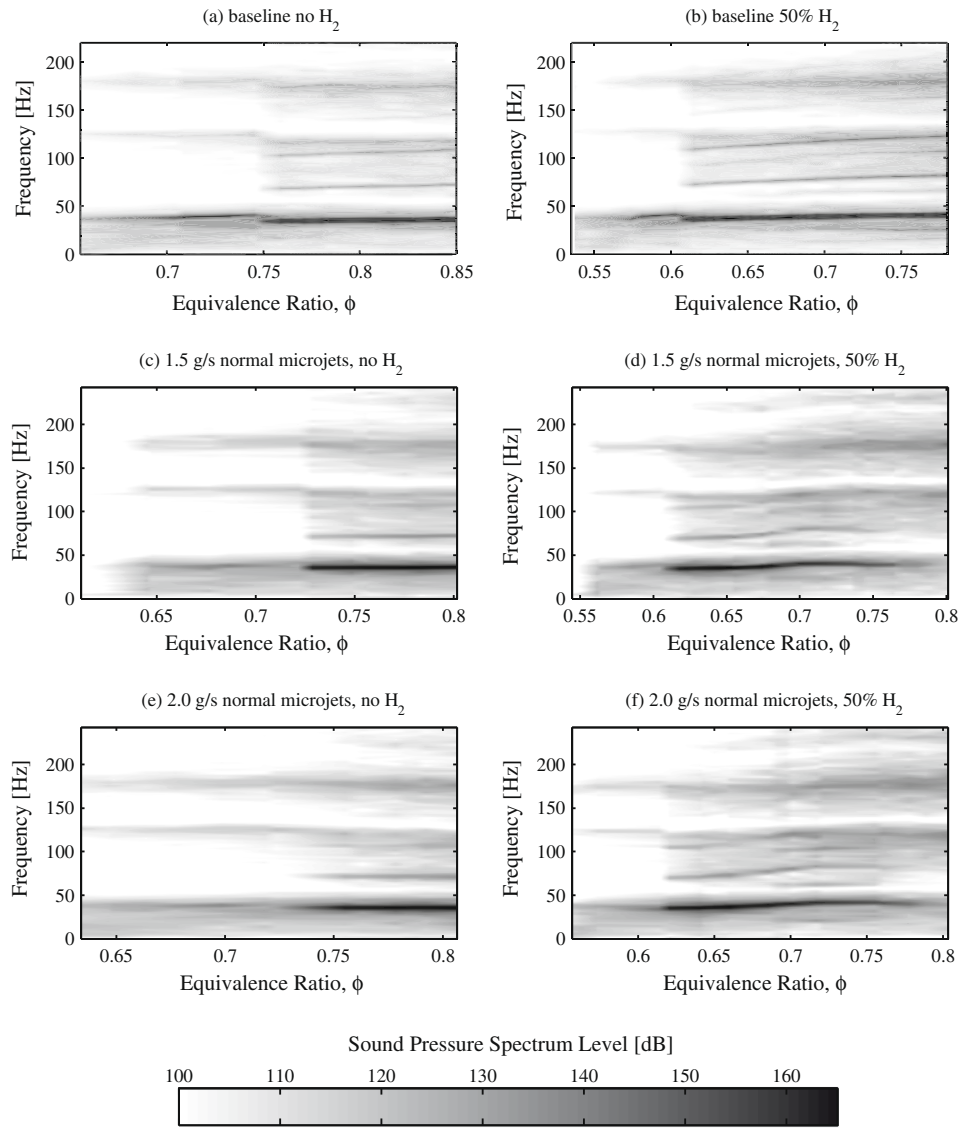

Fig. 7. Sound pressure spectrum level maps as a function of equivalence ratio for the same cases in Fig. 6.

jets, the excited higher acoustic mode resonates around 85 Hz, which is the 3/4 wavemode of the combustor. The response of the 7/4 wavemode around 200 Hz is also significant. The normal microjets damp/stabilize the acoustic mode around 132 Hz, as a result allowing the 85 Hz to be excited although in that case the flame is close to a pressure node as mentioned before. Since the excited resonant frequency is lower with normal microjets, the equivalence ratio at the transition is also lower compared to the baseline case. It is also interesting to observe that without hydrogen enrichment, the transition equivalence ratio is lower when the microjet flow rate is 1.5 g/s, compared to the microjet flow rate of 2.0 g/s. These observations will become more clear when we extend the theory explained in Section 3.3.3 to the 3/4 wavemode.

Next, we investigate the flame images recorded at different operating conditions and secondary air flow rates to examine the impact of normal microjets on the flame–vortex interaction mechanism.

3.3.2. Flame images

In order to distinguish between the dynamics at different operating conditions with passive control using normal microjets, we examine the flame images extracted from high-speed videos of different operating modes and describe corresponding unsteady pressure, and heat-release rate measurements.

First, a typical operating condition when the normal microjets are ineffective in stabilizing the flame is investigated. In Fig. 10, a sequence of flame images is shown, recorded at: $\phi = 0.57$, without hydrogen enrichment, the temperatures of the inlet mixture and the normal microjets are 600 K, and the secondary air flow rate is 1.50 g/s. Fig. 11 shows simultaneous heat-release rate and pressure measurements covering the same time period as the images shown in Fig. 10. The resonant frequency is 39 Hz, which corresponds to the 1/4 wavemode of the combustor.

The images show a typical unstable flame interacting with the large unsteady wake vortex formed at the recirculation zone behind the step [26,18] The wake vortex is formed (1–2) when the pressure is at its maximum. Since the combustor resonates at the 1/4 wavemode, the velocity oscillations lag the pressure oscillations by 90°; thus, this instant also corresponds to the moment of maximum flow acceleration. The wake vortex convects downstream towards the upper wall of the combustor as the pressure drops (2–5). As a result, a packet of unburned gases forms between the vortex and the flame of the previous cycle. In addition, the heat-release rate is decreasing following the end of the intense

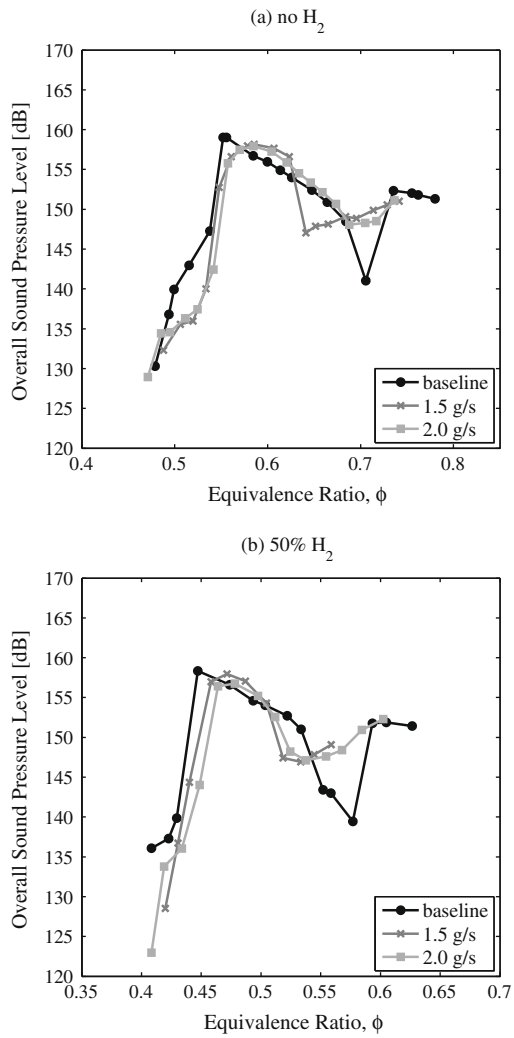

Fig. 8. OASPL as a function of equivalence ratio at different normal microjet flow rates (a) without hydrogen enrichment and (b) with 50% by volume hydrogen enrichment. The temperatures of the inlet mixture and the normal microjets is 600 K.

burning from the previous cycle. At instant 4, the heat-release rate reaches its minimum, and starts to rise as the unburned mixture trapped inside the vortex begin to burn (4–5). When the vortex reaches the upper wall of the combustor, the heat-release rate reaches its maximum (6–7) value, because the flame area is at its maximum value. Next, the heat-release rate starts to drop, fresh reactants enter the flame anchoring zone and the cycle repeats. Because at this inlet temperature the average velocity is high, flow never reverses, thus the flame never flashes back upstream of the step. The pressure and the heat-release rate are nearly in-phase resulting in the positive feedback between the unsteady heat-release rate and the pressure, driving the instability. The maximum pressure amplitude is near 1.8 kPa, and the OASPL is around 157 dB.

Next, we investigate the combustion dynamics when the resonant frequency is around 85 Hz corresponding to the 3/4 wavemode of the combustor. Note that the combustor does not couple with the 3/4 wavemode before activating the normal microjets. In Fig. 13, a sequence of flame images is shown, recorded at: $\phi = 0.72$, without hydrogen enrichment, the temperatures of the inlet mixture and the normal microjets are 600 K, and the secondary air flow rate is 1.50 g/s. Fig. 14 shows simultaneous heat-release rate and pressure measurements covering the same time period as the images shown in Fig. 13. In this figure, we plot the pressure oscillations measured at two different locations: (i) 0.15 m downstream of the choke plate and (ii) 1.28 m downstream of the choke plate, which is near the flame location. We observe that the pressure near the choke plate oscillates at 91 Hz, which corresponds to the 3/4 wavemode; whereas the pressure near the flame oscillates at 197 Hz, which corresponds to the 7/4 wavemode. This observation is verified in Fig. 12, where we plot the sound pressure spectra measured using both pressure sensors.

The heat-release rate measurements and the flame images show that the combustion dynamics are coupled with the 3/4 wavemode of the combustor, although the oscillation frequency near the flame corresponds to the 7/4 wavemode. The combustion dynamics captured in Fig. 13 is similar to the dynamics explained above. In this case, the flame is more compact, since it oscillates at a higher frequency. Near instant 1, the wake vortex is formed near the step, when the pressure is at its maximum. At this moment, the acceleration of the flow should also be maximum, suggesting that the velocity lags the pressure by 90° similar. This means that the flame is upstream of the pressure node. Because the pressure node is close to the flame for the 3/4 wavemode (see Fig. 5), the flame might also have been located downstream of the pressure node, which would cause the velocity to lead the pressure by 90°. In that case the vortex should have formed when the pressure was at its minimum value. The wake vortex convects downstream and expands towards the upper wall of the combustor (2–6), forming a reactant packet between the flame wrapped around the vortex, and the vertical flame formed in the previous cycle. Again, when the vortex touches the upper wall, the heat-release rate reaches its maximum value (7). At that moment, a new vortex forms at the step, and the cycle repeats. The pressure and the heat-release rate are in-phase, driving the instability.

Finally, we investigate the combustion dynamics for a case when normal microjets effectively suppress the instability. In Fig. 15, a sequence of flame images is shown, recorded at: $\phi = 0.80$, with 50% hydrogen enrichment, the temperatures of the inlet mixture and the normal microjets are 300 K, and the secondary air flow rate is 1.50 g/s. Fig. 16 shows simultaneous heat-release rate and pressure measurements covering the same time period as the images shown in Fig. 15. The images show that the dynamics are steady, there is no unsteady interaction between the flame and the unsteady wake vortex. The flame is anchored upstream of the step at the upper wall of the combustor. The heat-release rate is nearly steady and does not couple with the pressure oscillations. The pressure oscillation amplitude is around 0.4 kPa. At this operating condition, the burning velocity of the flame is high, and the mean flow velocity is low (because the inlet temperature is atmospheric). Without secondary air injection through normal microjets, at this operating condition the flame detaches from the recirculation zone and flashes back to the inlet channel. However, when the normal microjets are turned on, they prevent the flame from flashing back. Apparently, small scale vortices are generated as a result of the interaction of the microjets with the main flow, and as the microjets hit the upper wall of the combustor, creating a recirculation zone strong enough to anchor the flame at that location [27].

These results show that the normal microjets cannot prevent the formation of the unsteady vortex downstream of the step when the operating conditions correspond to the unstable or the high-frequency unstable modes. However, as the operating conditions are changed so that the burning velocity of the flame increases, the role of the unsteady vortex on the overall dynamics becomes less significant. That is why the OASPL starts to decrease at high equivalence ratios with hydrogen enrichment (see Fig. 6). Before activating the normal microjets, when the burning velocity of the flame exceeds a certain threshold, the flame detaches from the step

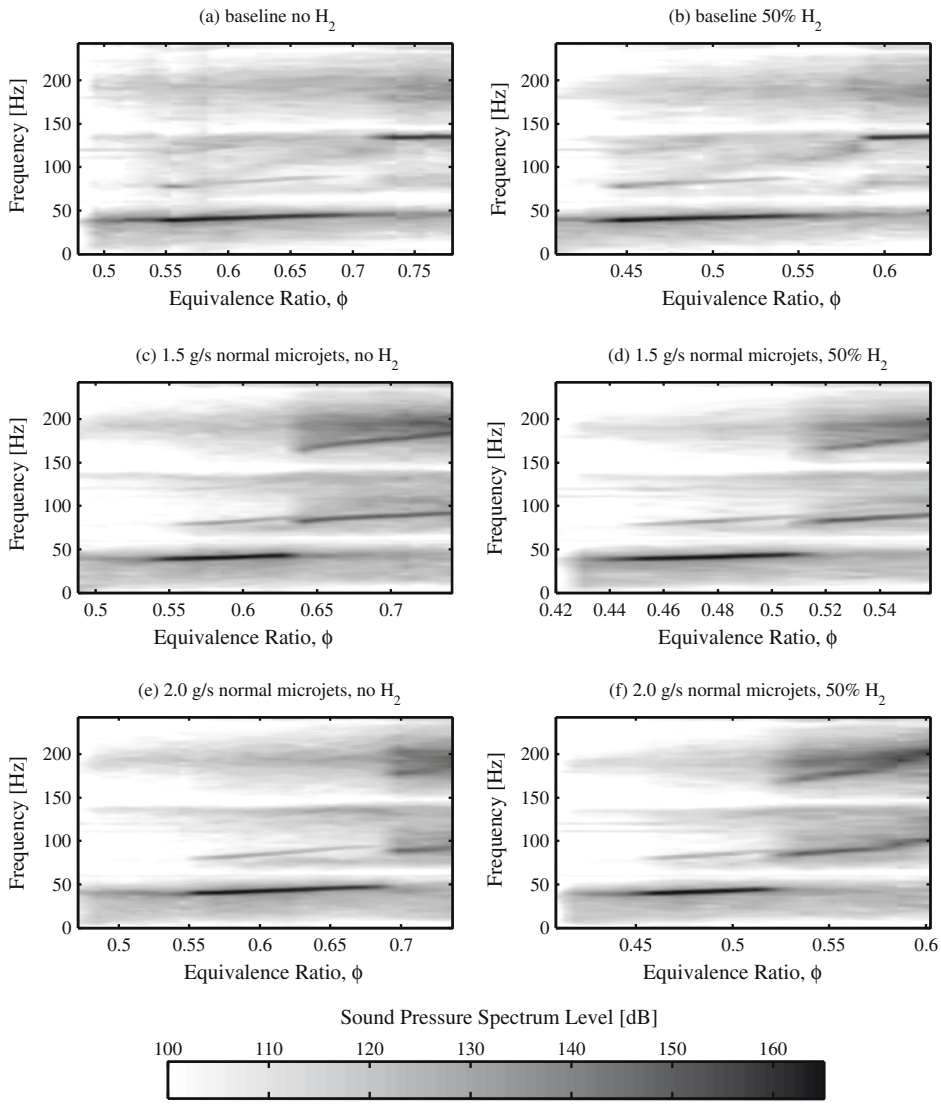

Fig. 9. Sound pressure spectrum level maps as a function of equivalence ratio for the same cases in Fig. 8.

and flashes back upstream. At these operating conditions, when the normal microjets are turned on, instead of the unsteady recirculation zone formed downstream of the step, the flame anchors upstream of the step in the "new, stable, aerodynamically created" intense recirculation zone formed near the location of microjets, towards the upper wall of the combustor. As a result, the flame dynamics are stabilized.

3.3.3. Model

In Ref. [18], we developed a model to predict the conditions at which the transition between different operating modes take place when the dynamics are driven by the flame–vortex interactions. When the normal microjets are introduced – except the operating conditions when they suppress the instability – the mechanism of flame–vortex interactions are still strong, and governs the combustion dynamics (see Figs. 10 and 13). Therefore, we expect this model to be also applicable for the cases with the normal microjets. The model developed in Ref. [18] is extended to predict the operating conditions at which a transition to the 3/4 wavemode resonating around 85 Hz is expected. For convenience, we summarize and generalize the model below.

The flame images show that when the wake vortex reaches the upper wall of the combustor, vortex breaks up and the heat-release rate reaches its maximum value. The vortex forms at the moment of maximum flow acceleration. We define the time elapsed between the moment of vortex formation and the moment when the vortex reaches the upper wall (moment of maximum heat-release rate) as τ_{vor}. We assume that the vortex expands towards the wall predominantly as a result of the expansion velocity in the vertical direction. The expansion velocity is proportional to the heat release parameter $S_T(\rho_u/\rho_b - 1)$, where S_T is the turbulent burning velocity and ρ_u and ρ_b are the unburned and burned mixture densities [28]. Assuming a linear relationship between the turbulent and laminar burning velocities, and taking into account of the strain rate on the burning velocity, the vertical velocity of the vortex, v_{vor}, is:

$$v_{vor} = \frac{S_c}{c}(\rho_u/\rho_b - 1) \tag{2}$$

where S_c is the strained laminar consumption speed of the flame. The constant of proportionality between the expansion velocity and the heat-release parameter, and the proportionality between the turbulent and laminar consumption speeds are represented by an arbitrary constant c. Representing the vertical distance between the corner of the step where the vortex forms and the upper wall as H, we obtain $\tau_{vor} = H/v_{vor}$. Substituting v_{vor} from Eq. (2):

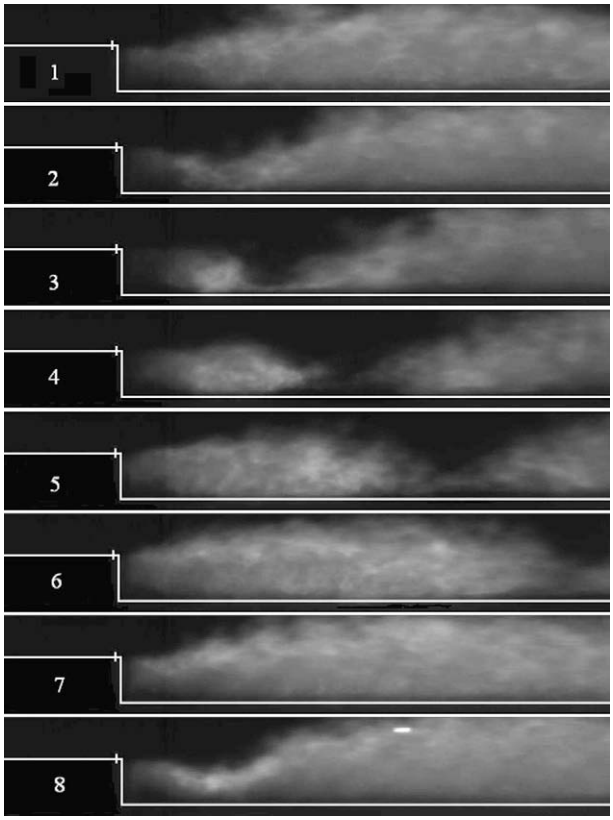

Fig. 10. Sequential flame images with 1.50 g/s normal microjets, without hydrogen enrichment, at $\phi = 0.57$. The temperatures of the inlet mixture and the normal microjets is 600 K. The time between the frames is 4.0 ms.

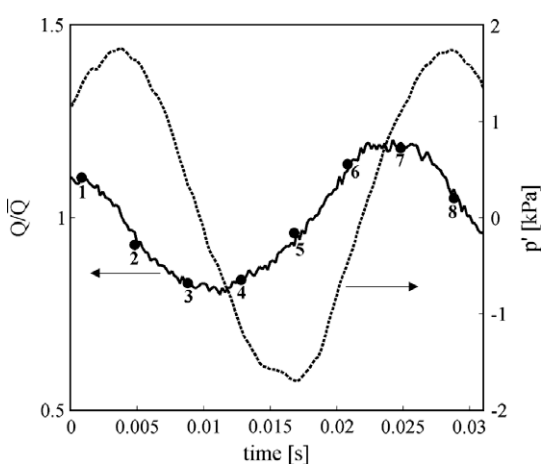

Fig. 11. Simultaneous pressure and heat-release rate measurements corresponding to the instants in Fig. 10.

$$\tau_{vor} = \frac{cH}{S_c(\rho_u/\rho_b - 1)} \tag{3}$$

The transitions occur when the pressure and heat-release rate oscillations are in-phase, that is $\tau_{pq} = NT$, where τ_{pq} is the time delay between the pressure and the heat-release rate oscillations, N is an integer and T is the oscillation period. Therefore, the transitions between the operating modes take place when (for $N = 1$):

$$\tau_{vor} = \alpha\frac{1}{f} \tag{4}$$

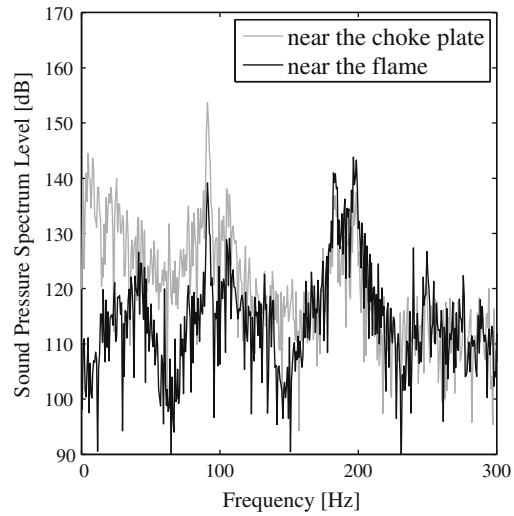

Fig. 12. Sound pressure spectrum levels measured near the choke plate and near the flame.

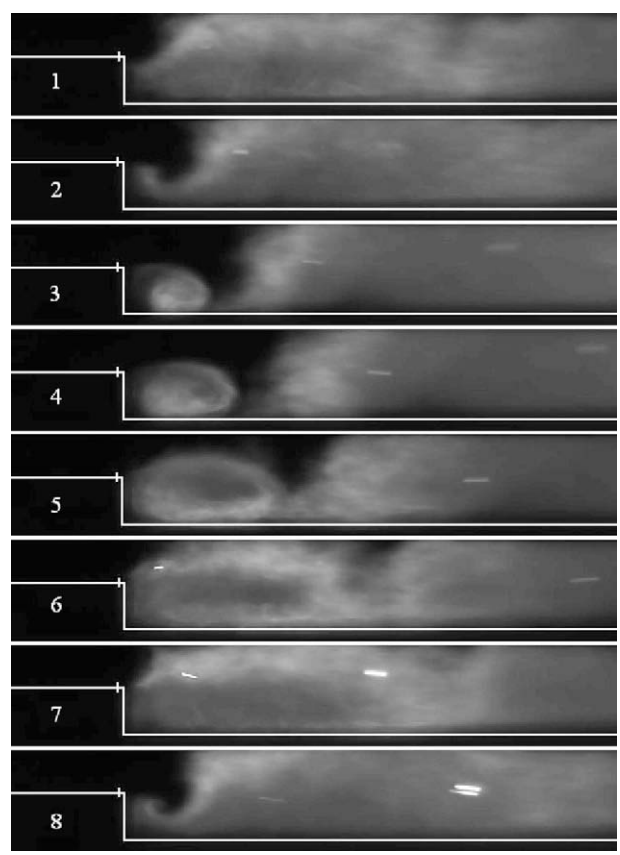

Fig. 13. Sequential flame images with 1.50 g/s normal microjets, without hydrogen enrichment, at $\phi = 0.72$. The temperatures of the inlet mixture and the normal microjets are 600 K. The time between the frames is 1.8 ms.

where $\alpha = 1$ if the velocity lags the pressure by 90° at the flame location, and $\alpha = 3/2$ if the velocity leads the pressure by 90° at the flame location. f is the resonant frequency. Using Eqs. (3) and (4), at the transitions between operating modes:

$$S_c(\rho_u/\rho_b - 1)|_{tr} = cHf/\alpha \tag{5}$$

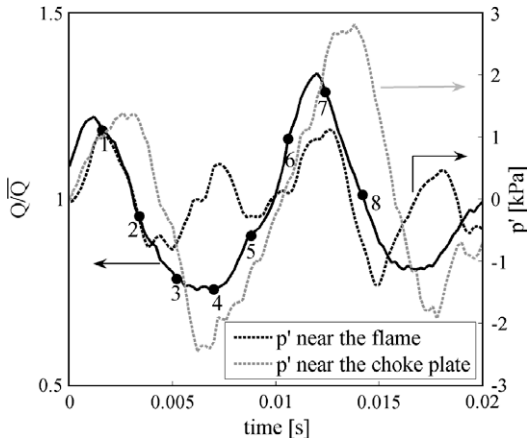

Fig. 14. Simultaneous pressure and heat-release rate measurements corresponding to the instants in Fig. 13.

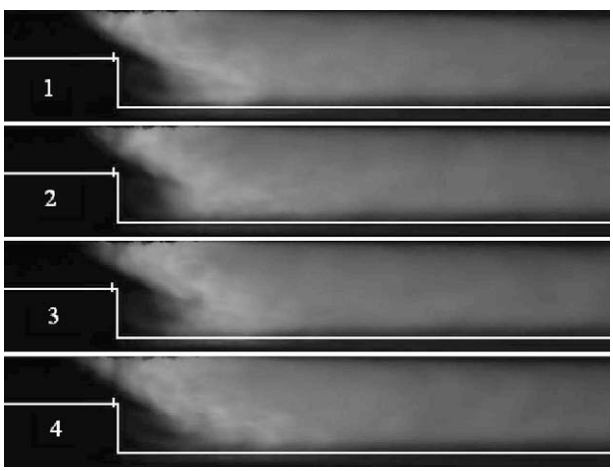

Fig. 15. Sequential flame images with 1.50 g/s normal microjets, with 50% by volume hydrogen enrichment, at $\phi = 0.80$. The temperatures of the inlet mixture and the normal microjets are 300 K. The time between the frames is 27 ms.

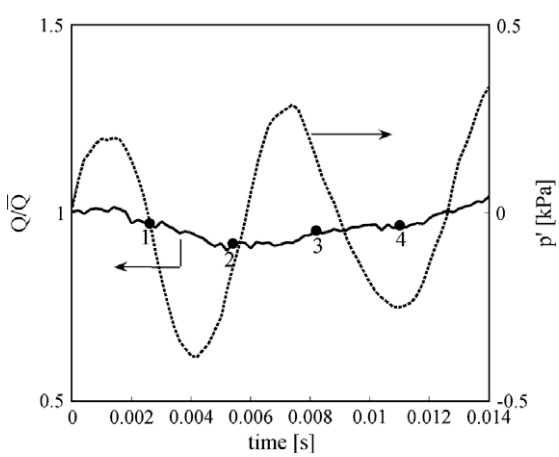

Fig. 16. Simultaneous pressure and heat-release rate measurements corresponding to the instants in Fig. 15.

The left-hand side of Eq. (5), i.e. the heat release parameter at transition, depends on the operating conditions (ϕ, fuel composition and T_{in}). For a set of operating conditions satisfying Eq. (5), the transitions between the operating modes are expected to take place.

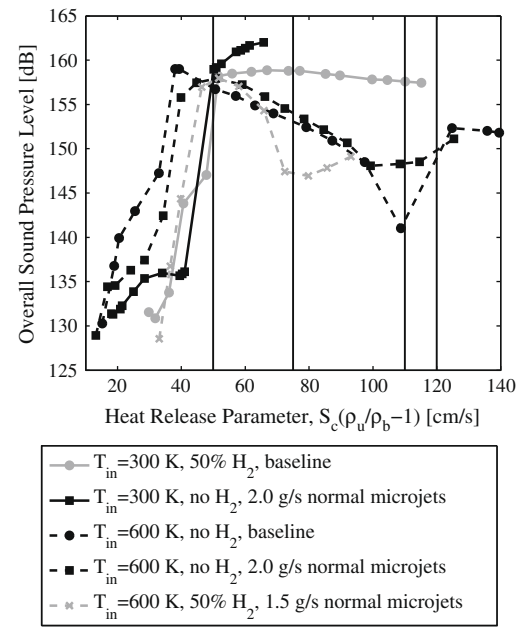

Fig. 17. OASPL as a function of heat release parameter, $S_c(\rho_u/\rho_b - 1)$, at different hydrogen concentrations, inlet temperatures and normal microjet flow rates.

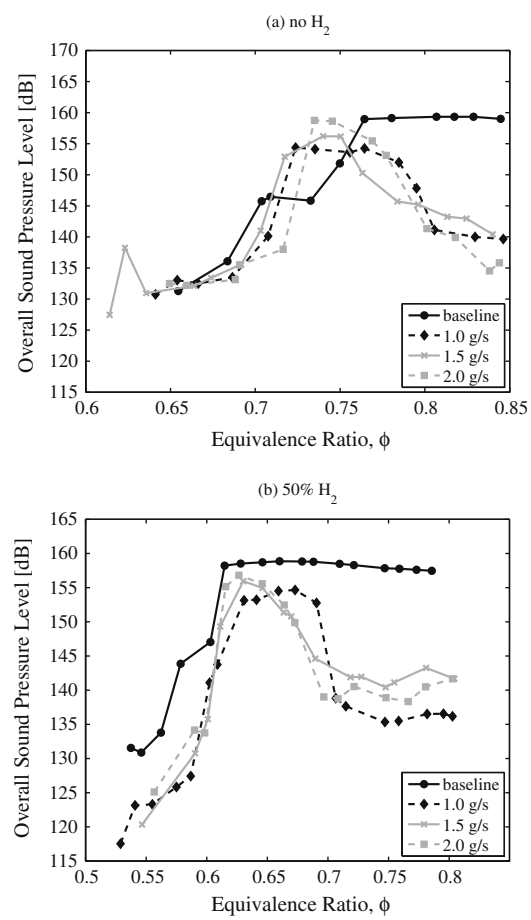

Fig. 18. OASPL as a function of equivalence ratio at different axial microjet flow rates (a) without hydrogen enrichment and (b) with 50% by volume hydrogen enrichment. The temperatures of the inlet mixture and the axial microjets are 300 K.

In Fig. 17, we plot the OASPL as a function of the heat release parameter, $S_c(\rho_u/\rho_b - 1)$, for arbitrarily selected cases from Figs. 6 and 8. The transition heat release parameters cannot be determined easily using the OASPLs measured by the pressure sensor near the flame, therefore we also plot the OASPLs measured using the pressure sensor near the choke plate. We use the strain rates of 2000 s^{-1} and 3000 s^{-1} for inlet temperatures of 300 K and 600 K, respectively, which give the best collapse of the data. The additional air introduced through the microjets is expected to decrease the overall equivalence ratio. However, when we calculate the overall equivalence ratio, the lean blowout limit is well below the values shown in Fig. 3, especially for the cases with $T_{in} = 300$ K. On the other hand, the blowout limits remain almost the same as the baseline cases with normal microjets when the equivalence ratio is calculated based only on the main air flow rate as shown in Figs. 6 and 8. This suggests that the secondary air does not mix with the reactant mixture completely at the burning zone. For this reason, the heat release parameter is calculated using the equivalence ratio based on the main air flow rate.

Fig. 17 shows that all the data are well correlated with the heat release parameter. The observed transitions between different operating modes occur at heat release parameters of around 50 cm/s, 75 cm/s, 110 cm/s and 120 cm/s. Note that the transition around 50 cm/s represents the transition to the unstable mode resonating at the 1/4 wavemode of the combustor. The transitions at 75 cm/s and 110 cm/s represent the excitation of the 3/4 wavemode, which was not excited without activating the normal microjets. Because the flame is very close to a pressure node when 3/4 wavemode is excited, the flame location might be upstream or downstream of the pressure node. If the flame is upstream/downstream of the pressure node, the velocity lags/leads the pressure by 90°. According to the theory represented in Eq. (5), the value of α is either 1 or 3/2, respectively, resulting in two distinct transitions when the 3/4 wavemode is excited. The transition at 120 cm/s represents the excitation of the 5/4 wavemode corresponding to the high-frequency unstable operating mode of the combustor without activating the normal microjets.

Using Eq. (5), the model predicts the ratio of the heat release parameters between the first 3/4 wavemode transition, and the quasi-stable to unstable transition as:

$$R_{3/4,1} = \frac{2}{3}\frac{f_{3/4,1}}{f_{1/4}} \tag{6}$$

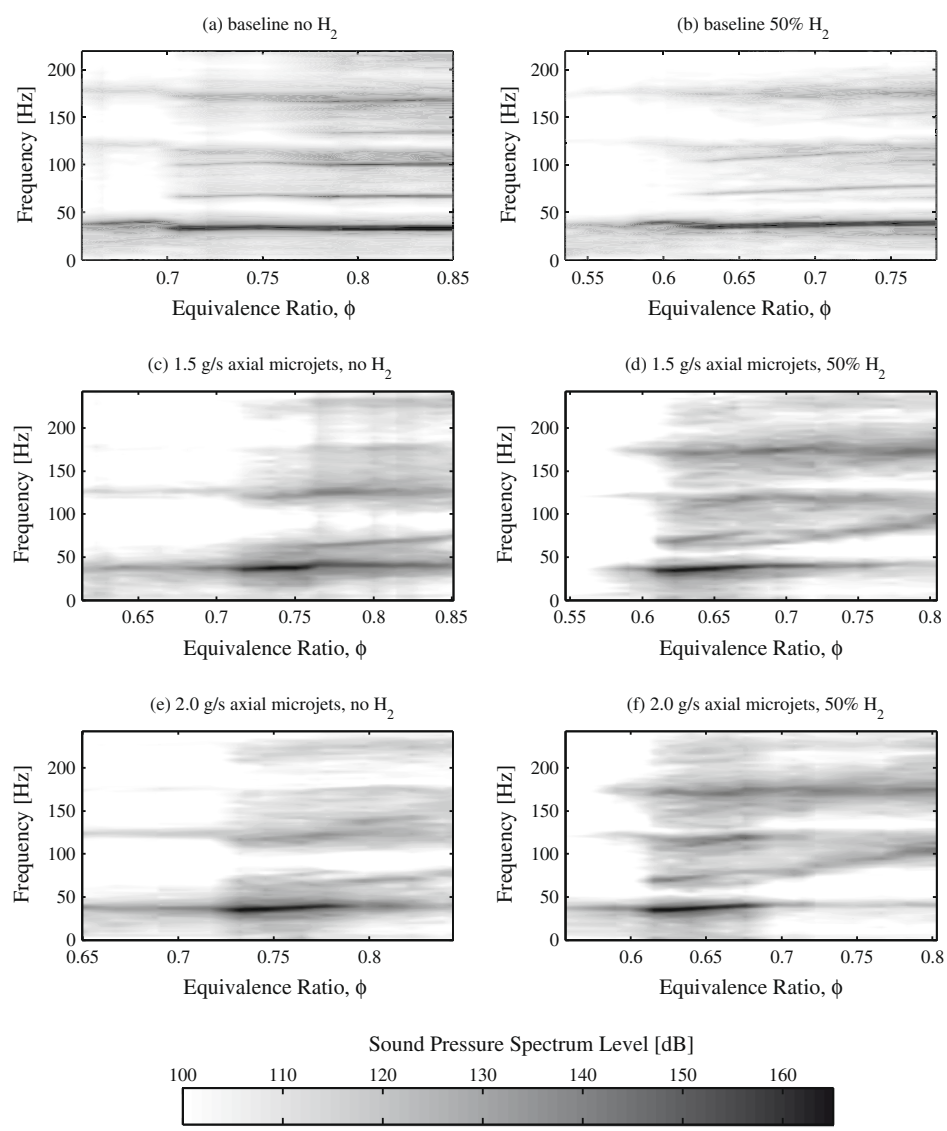

Fig. 19. Sound pressure spectrum level maps as a function of equivalence ratio for the baseline, 1.5 g/s and 2.0 g/s axial microjet flow rate cases in Fig. 18.

where $f_{3/4,1} = 85$ Hz and $f_{1/4} = 39$ Hz from Fig. 9c, d and f, giving $R_{3/4,1} = 1.45$. From the experimental data shown in Fig. 17, $R_{3/4,1} = 75/50 = 1.50$. Similarly, the ratio between the heat release parameters at the second 3/4 wavemode transition, and the quasi-stable to unstable transition is:

$$R_{3/4,2} = \frac{f_{3/4,2}}{f_{1/4}} \tag{7}$$

where $f_{3/4,2} = 89$ Hz and $f_{1/4} = 39$ Hz. Note that the 3/4 wavemode resonant frequency is slightly higher at the second transition as shown in Fig. 9e. The ratio is calculated as $R_{3/4,2} = 2.28$. From the experimental data shown in Fig. 17, $R_{3/4,2} = 110/50 = 2.20$.

Finally, the model predicts the ratio of the heat release parameters between the high-frequency unstable mode transition to the quasi-stable to unstable transition as:

$$R_{5/4} = \frac{2}{3}\frac{f_{5/4}}{f_{1/4}} \tag{8}$$

The resonant frequencies before activating the normal microjets, corresponding to high-frequency unstable and unstable cases, respectively, are $f_{5/4} = 132$ Hz and $f_{1/4} = 36$ Hz as mentioned in Section 3.1. Note that without normal microjets the value of $f_{1/4}$ is lower. The model gives $R_{5/4} = 2.44$. From the experimental measurements $R_{5/4} = 120/50 = 2.40$.

The model predicts all the transitions with good accuracy. It is applicable as long as the dynamics are governed by the flame–vortex interaction mechanism.

3.4. Combustion dynamics with axial microjets

We also performed experiments to investigate the effect of secondary air injection through axial microjets on the combustion dynamics. We performed pressure measurements and examined the flame images at different fuel compositions, inlet temperatures, equivalence ratios and microjet flow rates.

3.4.1. Pressure measurements

In Fig. 18, we plot the OASPL as a function of the equivalence ratio based on the main air flow rate at different axial microjet flow rates without hydrogen enrichment and with 50% by volume hydrogen enrichment. The temperatures of the inlet mixture and the secondary air flow are 300 K. The axial microjets destabilize the quasi-stable flame without hydrogen enrichment, whereas they stabilize the quasi-stable flame when 50% by volume hydrogen is added to propane. The stable flame is not impacted by the axial microjets without hydrogen enrichment, but the OASPL decreases around 7 dB in the stable regime with 50% by volume hydrogen enrichment. The OASPL drops across the entire range of equivalence ratio, however the reduction in the OASPL is maximum and around 20 dB when $\phi > 0.80$ without hydrogen enrichment and $\phi > 0.70$ with 50% by volume hydrogen enrichment. The impact of the axial microjet flow rate on the combustion dynamics is not strong.

Fig. 19 shows the combustor's frequency response corresponding to the baseline, 1.5 g/s and 2.0 g/s axial microjet flow rate cases in Fig. 18. The 1/4 wavemode resonating around 40 Hz is dominant except for the cases when the axial microjets stabilize the unstable dynamics. When the axial microjets stabilize the dynamics, the sound pressure level at the resonant frequency decreases substantially; on the other hand, the sound pressure level at the 175 Hz becomes stronger. The baseline cases show that the resonant frequency slightly increases in the quasi-stable regime compared to the unstable regime, causing a discontinuity in the spectrum maps. However, with axial microjets, without hydrogen enrichment, the unstable band is continuous until the stable band is reached. With 50% by volume hydrogen enrichment, the quasi-stable band completely disappears, causing a sudden transition from unstable to stable dynamics.

Next, we increased the temperatures of the inlet mixture and the secondary air to 600 K. Fig. 20 shows the OASPL as a function of the equivalence ratio based on the main air flow rate at different axial microjet flow rates without hydrogen enrichment or with 50% by volume hydrogen enrichment. Although the axial microjets at a flow rate of 1 g/s do not suppress the pressure oscillations except at a narrow region near the quasi-stable to unstable transition without hydrogen enrichment, they suppress the pressure oscillations by around 20 dB at the entire equivalence ratio range corresponding to unstable and high-frequency unstable operating modes when 50% by volume hydrogen is added to propane. When the axial microjet flow rate is increased to 1.5 g/s, the pressure oscillations are suppressed by 15 dB, both without and with hydrogen enrichment except the region near the quasi-stable to unstable region. As the microjet flow rate is increased further to 2 g/s, the high-frequency unstable flame can no longer be stabilized. These results show that there is an optimum axial microjet flow rate depending on the fuel composition and the inlet temperature at which a robust stabilization of the dynamics is achieved.

In Fig. 21, we plot the combustor's frequency response corresponding to the baseline, 1.5 g/s and 2.0 g/s axial microjet flow rate cases in Fig. 20. The introduction of axial microjets cause the resonant frequency to shift to 85 Hz, corresponding to the 3/4 wavemode at intermediate values of equivalence ratios, except at the

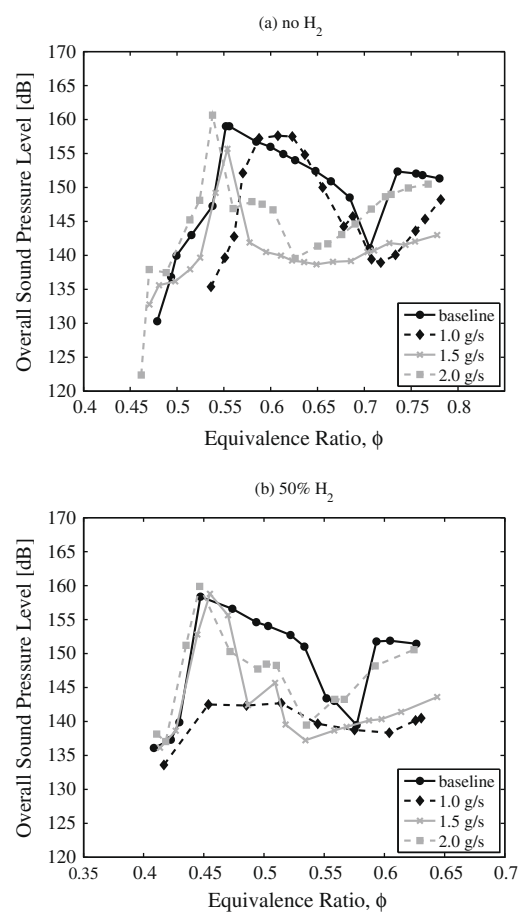

Fig. 20. OASPL as a function of equivalence ratio at different axial microjet flow rates (a) without hydrogen enrichment and (b) with 50% by volume hydrogen enrichment. The temperatures of the inlet mixture and the axial microjets are 600 K.

axial microjet flow rate of 1.5 g/s, without hydrogen enrichment. At this operating condition, the amplitude of the oscillations at the 85 Hz mode is always lower than that at the 40 Hz mode. When the axial microjet flow rate is 1.5 g/s, the high-frequency unstable mode is suppressed. As the axial microjet flow rate is increased to 2.0 g/s, the 3/4 wavemode is excited independent of the hydrogen concentration in the fuel, and the high-frequency unstable mode cannot be suppressed, suggested by the dark bands observed at high equivalence ratios at both hydrogen concentrations. The excitation of the 3/4 wavemode is also suggested by the slight jumps in the OASPL in Fig. 20.

3.4.2. Flame images

In order to distinguish between the dynamics at different operating conditions with passive control using axial microjets, we examine the flame images extracted from high-speed videos of different operating modes and describe corresponding unsteady pressure, and heat-release rate measurements.

First, we investigate a typical operating condition when the axial microjets are ineffective in stabilizing the flame. In Fig. 22, a sequence of flame images is shown, recorded at: $\phi = 0.55$, without hydrogen enrichment, the temperatures of the inlet mixture and the axial microjets are 600 K, and the secondary air flow rate is 1.50 g/s. Fig. 23 shows simultaneous heat-release rate and pressure measurements covering the same time period as the images shown in Fig. 22. The resonant frequency is 37 Hz, which corresponds to the 1/4 wavemode of the combustor. In this case, the secondary air injection through axial microjets causes the flame at the recirculation zone to blow out. Therefore, the flame anchors near the hot top wall of the combustor. The formation of the wake vortex, and its convection in the downstream direction towards the upper wall of the combustor cannot be observed from these images.

Close to instant 1, the pressure is at its maximum value, i.e. the flow acceleration is maximum (recalling that the pressure leads the velocity by 90° when the resonant frequency corresponds to the 1/4 wavemode). At this moment, we expect a wake vortex to be formed in the recirculation zone. The flame is pushed downstream with the high flow velocity, while the vortex convects downstream towards the upper wall of the combustor (2–4). As the inlet velocity drops below its mean value, the flame near the upper wall of the combustor moves towards the step (5). At this moment, the flame area, thus the heat-release rate start to rise. At instant 6, the flame reaches the step and is wrapped around

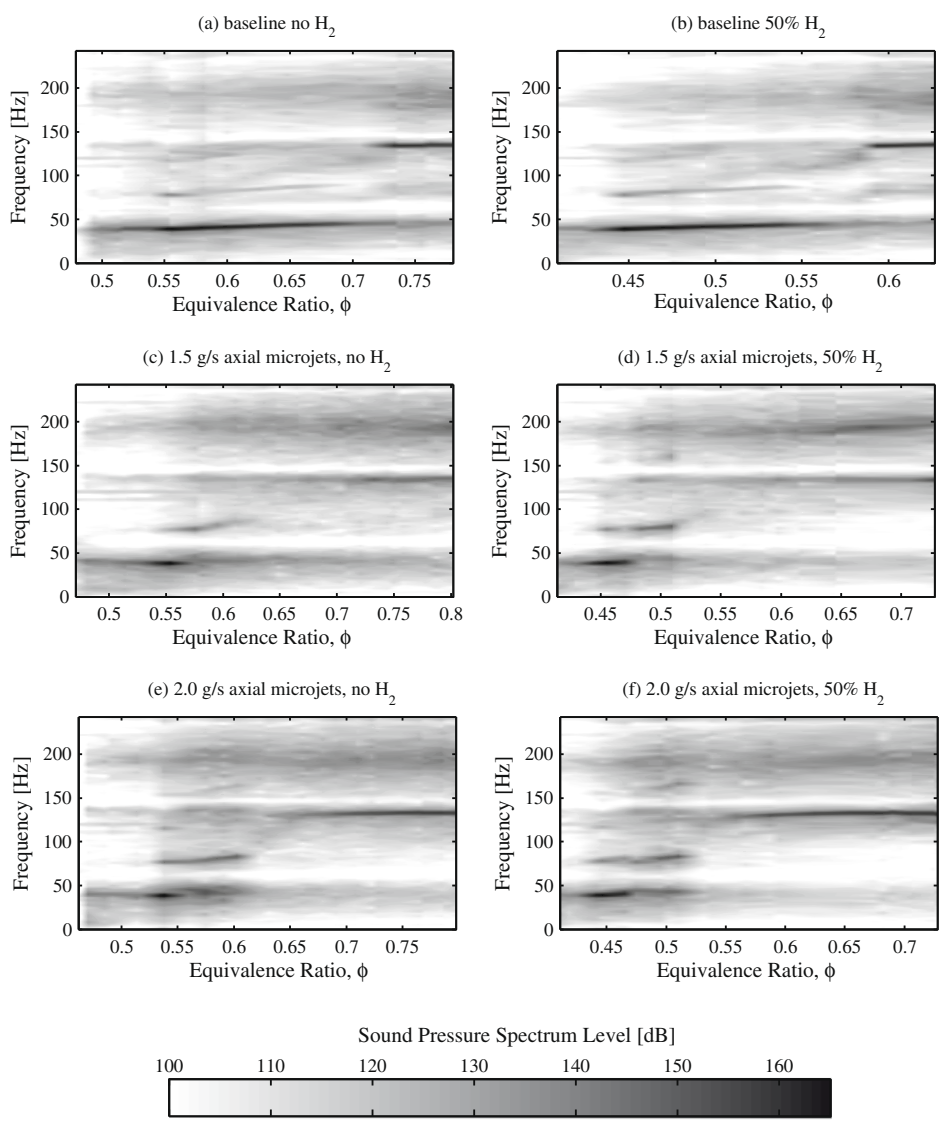

Fig. 21. Sound pressure spectrum level maps as a function of equivalence ratio for the baseline, 1.5 g/s and 2.0 g/s axial microjet flow rate cases in Fig. 20.

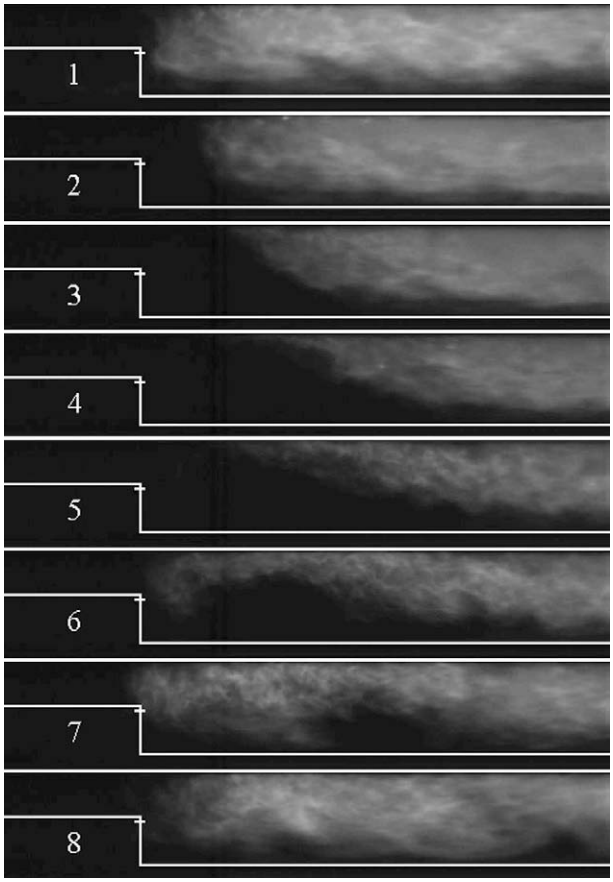

Fig. 22. Sequential flame images with 1.50 g/s axial microjets, without hydrogen enrichment, at $\phi = 0.55$. The temperatures of the inlet mixture and the normal microjets are 600 K. The time between the frames is 3.9 ms.

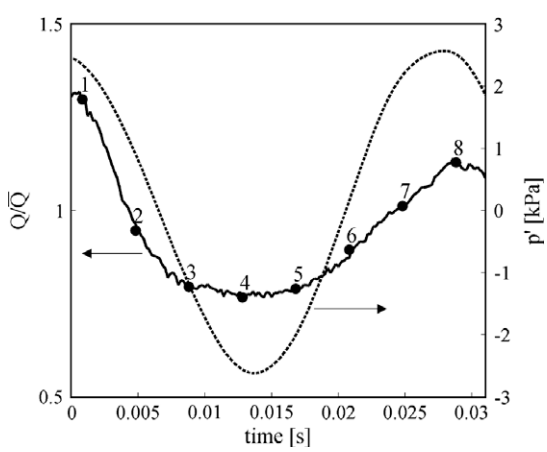

Fig. 23. Simultaneous pressure and heat-release rate measurements corresponding to the instants in Fig. 22.

the vortex. The heat-release rate reaches its maximum value when the vortex reaches the upper wall of the combustor, and breaks down (8). The reactants within the vortex burn, generating significant flame area. The heat-release rate is in-phase with the pressure oscillations, generating thermoacoustic instability. The maximum pressure amplitude is around 2.6 kPa, close to the value measured during the unstable operating mode of the combustor. In this case, secondary air injection through axial microjets does not prevent the strong unsteady flame–vortex interaction mechanism.

Now, we examine the flame images when the axial microjets are effective in suppressing the unstable operating mode of the combustor. In Fig. 24, a sequence of flame images is shown, recorded at: $\phi = 0.66$, without hydrogen enrichment, the temperatures of the inlet mixture and the axial microjets are 600 K, and the secondary air flow rate is 1.50 g/s. Fig. 25 shows simultaneous heat-release rate and pressure measurements covering the same time period as the images shown in Fig. 24. At this equivalence ratio, the axial microjets do not cause the flame to blow out near the recirculation zone. The vortex at the recirculation zone is steady, and trapped within a small zone downstream of the step. The axial microjets act as a restriction, not allowing the vortex to convect downstream and move towards the upper wall of the combustor, preventing the flame–vortex interactions. Thus, the flame is steady, generating almost steady heat-release rate.

Finally, we examine the flame images when the axial microjets are effective in suppressing the high-frequency unstable operating mode of the combustor. In Fig. 26, a sequence of flame images is shown, recorded at: $\phi = 0.78$, without hydrogen enrichment, the temperatures of the inlet mixture and the axial microjets are 600 K, and the secondary air flow rate is 1.50 g/s. Fig. 27 shows simultaneous heat-release rate and pressure measurements covering the same time period as the images shown in Fig. 26. The flame shapes and the combustion dynamics with axial microjets are the same as above. Again, the unsteady interactions between the wake vortex and the flame are prevented by the axial microjets by which the hydrodynamics of the flame anchoring zone are stabilized. The

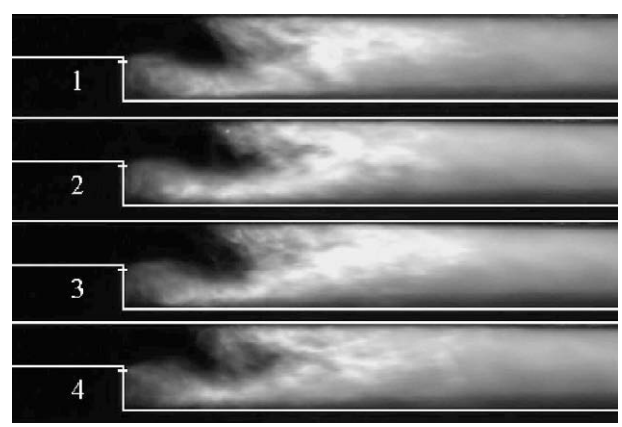

Fig. 24. Sequential flame images with 1.50 g/s axial microjets, without hydrogen enrichment, at $\phi = 0.66$. The temperatures of the inlet mixture and the axial microjets are 600 K. The time between the frames is 8.3 ms.

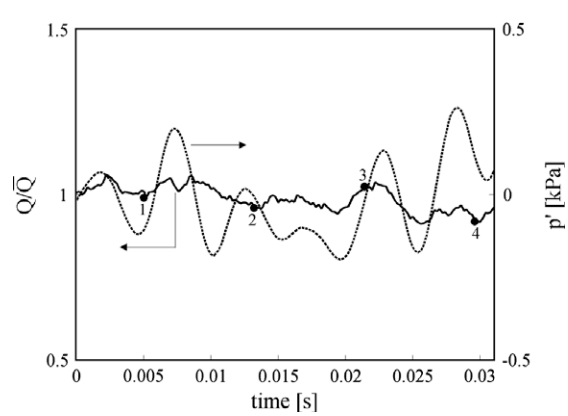

Fig. 25. Simultaneous pressure and heat-release rate measurements corresponding to the instants in Fig. 24.

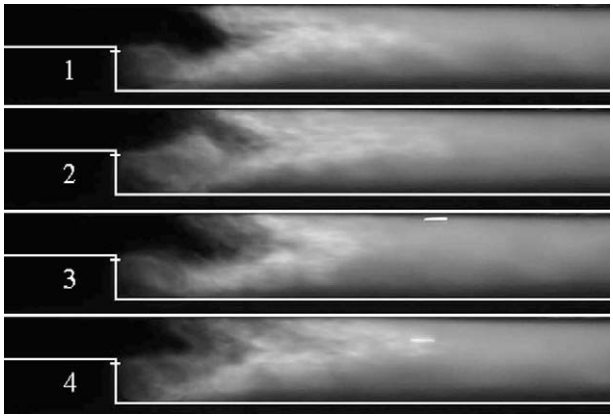

Fig. 26. Sequential flame images with 1.50 g/s axial microjets, without hydrogen enrichment, at $\phi = 0.78$. The temperatures of the inlet mixture and the axial microjets are 600 K. The time between the frames is 6.7 ms.

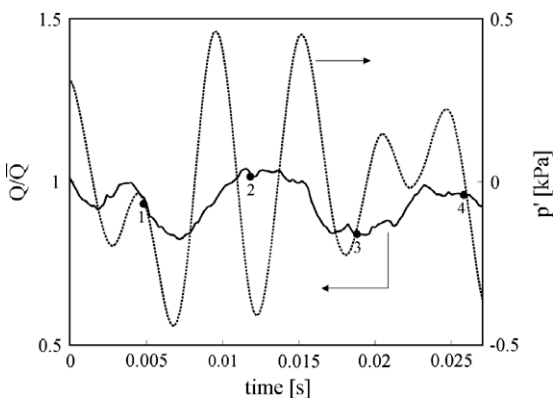

Fig. 27. Simultaneous pressure and heat-release rate measurements corresponding to the instants in Fig. 26.

heat-release rate is almost steady, does not couple with the acoustics of the combustor; thus preventing thermoacoustic instability.

In summary, after the axial microjets are turned on at an optimum axial microjet flow rate (i) preventing the local flame blowout near the step and (ii) high enough to impact the dynamics of the recirculation zone, the disruption of the flame–vortex interactions results in mismatch between the pressure-heat-release rate phase correlation, significantly reducing the pressure amplitude. As a result, the phase relationship between the pressure and the heat-release becomes random, leading to the stable dynamics [29].

4. Conclusions

In this paper, we experimentally investigated the effectiveness of passive control by injecting steady air flow near the step in the cross-stream or streamwise directions through choked microdiameter holes.

At operating conditions when the flame detaches from the step and flashes back into the upstream channel without air injection, when air was injected in the cross-stream direction, the flame anchors slightly upstream of the step in the "new, stable, aerodynamically created" intense recirculation zone formed near the upper wall of the combustor, instead of on the unsteady recirculation zone forming downstream of the step, which suppresses the instability. On the other hand, at operating conditions corresponding to the low-frequency unstable and high-frequency unstable operating modes of the combustor without air injection, air injection in the cross-stream direction is ineffective in stabilizing the dynamics, which is still governed by the flame–vortex interactions. In these cases, the 3/4 wavemode of the combustor, in addition to the 1/4 and the 5/4 wavemodes is excited. We demonstrated that the consumption speed of strained flames can be used to collapse the response curves describing the transitions among different dynamic modes onto a function of the heat release parameter alone. We predict the values of the heat release parameters at which transitions to the 3/4 and 5/4 wavemodes take place.

When air is injected in the streamwise direction near the edge of step, both the unstable and the high-frequency unstable dynamics could be stabilized at an optimum secondary air flow rate which depends on the operating conditions. When effective, the secondary air flow suppresses the shedding of an unsteady vortex, and hence the flame–vortex interaction mechanism. Instead, a compact, stable flame is formed near the step. On the other hand, when air injection in the streamwise direction is ineffective, the flame blows out near the step, anchoring at the hot top wall of the combustor. In this case, the flame is unsteady, the flame–vortex interactions are strong, and the dynamics are distinct compared to that at the low-frequency unstable and high-frequency unstable modes.

These results suggest that the interaction between the flame and the unsteady vortex shedding is a leading cause of the instability, as proposed earlier. They also suggest that blocking this interaction can lead to suppressing the dynamics.

In future studies, we plan to extend this work by performing high speed PIV measurements in order to determine the instantaneous velocity fields.

Acknowledgments

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